

Multimodal Fashion & Context Retrieval

Technical Report

This report documents the approaches considered, the chosen architecture, the dataset curation process (including issues found and how they were resolved), implementation challenges encountered, evaluation results against the five required queries, and directions for future work.

1. Approaches Considered

Five approaches were evaluated before settling on a final architecture. The core challenge flagged by the assignment is that vanilla CLIP retrieval struggles with compositionality (binding an attribute like a color to the correct garment, rather than treating a caption as an unordered bag of concepts) and with fine-grained fashion attributes. Each approach below was evaluated against how directly it addresses this, and at what cost.

Approach	How it works	Strengths	Weaknesses
1. Vanilla CLIP	Embed whole images and query text with off-the-shelf CLIP; cosine similarity search.	Simple, zero training, fast to build, genuinely zero-shot.	Fails on compositionality ("red shirt blue pants" vs. reverse); weak on fine-grained fashion vocabulary.
2. Structured attribute matching	Tag each garment with structured fields (category, color, attributes) via a fashion taxonomy; parse the query into the same slots; match symbolically.	Directly solves compositionality via explicit binding; interpretable.	Brittle outside the fixed taxonomy; not zero-shot for novel attribute words; heavier engineering.
3. Region-aware CLIP retrieval (chosen)	Crop each garment instance (using existing segmentation masks/bboxes); embed each crop plus the whole image separately; decompose queries into sub-phrases matched to the right region type.	Fixes compositionality by binding color/attribute to a specific crop, not the whole image; stays fully zero-shot; scales via standard vector search.	Depends on segmentation/crop quality; more vectors per image (storage cost); no benefit for images without masks.
4. Fine-tuned / hard-negative CLIP	Contrastively fine-tune CLIP (or use FashionCLIP) with attribute-swapped hard negatives to directly teach compositional understanding.	Could yield the best raw embedding quality; addresses the root cause in the model itself.	Requires training data, compute, and time; risk of overfitting to seen attribute combinations; least zero-shot in spirit.

5. Two-stage: retrieval + VLM rerank	Use fast CLIP/region retrieval for top-N candidates, then have a VLM read each candidate and re-score it against the literal query.	Best likely accuracy on compositional queries; VLM reasons over the whole image directly.	Added latency and a second model dependency; cost bounded but non-zero.
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Decision: Approach 3 (region-aware CLIP retrieval) was selected as the primary architecture. It directly targets the specific weakness the assignment calls out, requires no training data or fine-tuning (preserving zero-shot capability), and stays within a standard vector-database paradigm rather than requiring custom infrastructure in line with the assignment's instruction to focus effort on ML logic rather than engineering. Approach 5 (VLM reranking) was identified as a natural extension and is discussed further in Section 6 (Future Work).

2. Chosen Approach: Region-Aware CLIP Retrieval

2.1 Indexer

For every image with available segmentation masks, each main-garment instance (e.g. shirt, dress, pants, jacket excluding sub-garment “parts” such as sleeve, collar, or pocket, which are not meaningful standalone search targets) is cropped using its bounding box and embedded separately with CLIP. Every image, regardless of whether it has masks, additionally receives one whole-image embedding, used both for scene/context queries and as a fallback signal. All vectors are L2-normalized and stored in a FAISS flat index, with parallel metadata recording each vector's source image, region type, garment category, and detected color.

2.2 Retriever

An incoming query is decomposed into garment specific sub phrases (each optionally paired with a color, matched via keyword lookup against known Fashionpedia categories and a fixed color palette) plus a scene phrase, which is the query as a whole. Each garment sub-phrase is embedded and searched only against vectors of the matching region type, so “red tie” is compared exclusively to tie crops, not to the whole image. The scene phrase is compared against every image's whole-image vector. This separation is the mechanism that fixes compositionality: color and garment are checked independently per region, rather than blurred into one embedding the way vanilla CLIP would.

2.3 Handling images without segmentation masks

Roughly 18% of the final dataset (the Pexels-sourced supplement) has no segmentation masks and therefore no per-garment region vectors. For these images, a garment sub-phrase falls back to being compared against the whole-image vector instead, at a 0.8x confidence discount, only when no true region match exists for that image. Per-image scores across all matched sub-phrases are then averaged, not summed, so images with more regions (Fashionpedia) are not unfairly favored over images with only a scene vector (Pexels) purely because they contributed more matched terms.

2.4 Why this handles fashion queries specifically

- **Attribute-specific queries** (“bright yellow raincoat”): a single garment-color pair, matched directly against the relevant garment-category region vectors.

- **Contextual/place queries** (“inside a modern office”) : handled by the whole-image scene vector, independent of any garment matching.
- **Compositional queries** (“red tie and white shirt”) : the core case this architecture was built for: each garment-color pair is checked against its own crop, so the model cannot confuse which color belongs to which item.
- **Style-inference queries** (“casual weekend outfit”) : no explicit garment keyword to decompose, so retrieval relies on the whole-image embedding's general style/aesthetic understanding, which CLIP handles reasonably well.

3. Dataset Methodology

3.1 Source and curation

The dataset is built from **Fashionpedia** (validation split), which provides segmentation masks and category annotations for ~48,000 fashion images. After filtering to only annotated images (1,158 of 3,200 files in the validation package), the following issues were identified and addressed:

- **No color annotations exist in Fashionpedia's ontology** (294 attributes span fit, length, pattern, and construction but never color). Dominant color per garment instance was derived independently via k-means clustering on masked pixels, mapped to a fixed fashion-relevant color palette. This was used for dataset curation/reporting only; the retrieval pipeline itself relies on CLIP's own zero-shot color understanding.
- **Severe color imbalance**: 89.5% of all garment instances were black, beige, navy, brown, gray, or white. Since evaluation query 1 specifically requires yellow, the final 800-image curated subset preserves *all* instances of rare/saturated colors (yellow, orange, red, blue, green, purple, teal, pink, maroon) and fills remaining slots with a category-balanced sample of the rest.
- **No environment/location labels exist** in Fashionpedia at all. Zero-shot scene classification (five candidate labels: office, street, park, home, studio) was run over the curated subset. Results: studio 62.4%, street 28.1%, park 7.4%, home 1.1%, office 1.0%.

3.2 Supplementing office and home scenes

With office and home scenes each under 1.5% of the curated subset, evaluation query 2 (“business attire inside a modern office”) would have had almost no ground-truth matches to retrieve. 180 supplementary images (90 office, 90 home) were sourced from the Pexels API under the Pexels License. These images, lacking segmentation masks, were merged with the curated Fashionpedia subset into a single unified manifest of 980 images. Final scene distribution: studio 50.9%, street 23.0%, home 10.1%, office 10.0%, park 6.0%.

4. Implementation Challenges & Debugging

This section is included deliberately: understanding a chosen approach's shortcomings, including engineering ones encountered along the way, was an explicit evaluation criterion.

4.1 SigLIP-2 checkpoint mismatch

SigLIP-2 was the original choice of embedding model (sigmoid loss generally outperforms CLIP's contrastive loss on fine-grained retrieval). However, loading `google/siglip2-base-patch16-224` via `transformers 5.14.0` produced a confirmed upstream bug: the checkpoint loaded under the SigLIP v1 architecture class rather than `Siglip2Model`, causing tensor shape mismatches (position embeddings: 196 vs. 256) and a fatal load error. This is a known, currently open issue in the transformers repository, not a configuration mistake. After a bounded debugging effort, the decision was made to fall back to CLIP rather than keep sinking time into an unresolved upstream bug.

4.2 transformers v5 API change

A second, unrelated issue surfaced after switching to CLIP: `get_text_features / get_image_features` raised an `AttributeError` when normalizing the output. This was traced to a documented, intentional breaking change in transformers v5 as of v5, these methods return a `BaseModelOutputWithPooling` object for every model (not just SigLIP), rather than a plain tensor. The fix was to extract `.pooler_output` explicitly before normalizing.

4.3 Outcome

With both issues resolved, CLIP (`openai/clip-vit-base-patch32`) produced well-differentiated similarity scores (0.25–0.31 for top matches, versus a near-uniform 0.08–0.13 range before the fix, which indicated broken embeddings rather than genuine low relevance). SigLIP-2 remains a documented future-work upgrade once the upstream checkpoint issue is resolved.

5. Evaluation Results

Each of the five required queries was run against the built index and manually inspected. No ground-truth relevance labels exist for these queries (none were provided, and building them was out of scope), so results below reflect direct visual inspection of top-10 retrievals rather than a computed accuracy metric.

Query	Result	Finding
1. "Bright yellow raincoat"	Strong	Top result was a genuine bright-yellow raincoat/trench. Precision dropped after rank 1 CLIP conflates yellow with adjacent warm hues (mustard, tan, camel, orange). One result (rank 9) appears to have scored highly due to a yellow object in the background rather than the garment itself, a residual compositionality gap worth further investigation.
2. "Business attire inside a modern office"	Strong	This is the query most dependent on the Pexels office supplement and the whole-image fallback path, since Fashionpedia itself has almost no office imagery with garment masks.

3. “Blue shirt sitting on a park bench”	Mixed	The “blue shirt” attribute matched reasonably. The specific action/pose, sitting on a bench was largely ignored; results skewed toward general outdoor/park imagery rather than the specific bench-sitting pose. This is an expected CLIP limitation: strong on objects/colors/scenes, weak on fine-grained actions or poses.
4. “Casual weekend outfit for a city walk”	Strong	All top-10 results were street-style photos of casual outfits, jackets, jeans, sneakers. This query has no explicit garment keyword, so it ran almost entirely on the whole-image scene embedding, and CLIP's general style/aesthetic matching handled it well.
5. “Red tie and white shirt in a formal setting”	Architecture validated, data-limited	The “white shirt” component matched well across nearly all top-10 results. The “red tie” component was largely absent from top results, traced to data sparsity: only 3 tie instances exist in the entire 980-image dataset, so the tie sub-phrase search had almost nothing to retrieve against. This is a documented data limitation, not an architecture failure, the region-splitting mechanism itself worked as designed for the shirt component.

5.1 Honest summary

The architecture clearly outperforms a naive whole-image baseline on the queries it was specifically designed for (compositional attribute binding, query 5's shirt component; single-attribute matching, query 1). It performs well on broad style/aesthetic queries (query 4) where CLIP's general embedding is already strong. Its weakest points are: (a) fine-grained color precision within a hue family, (b) action/pose understanding (query 3), and (c) categories with very few training instances (ties, query 5), the last of which is a dataset limitation rather than an architectural one.

6. Future Work

6.1 Extending to location and weather

- **Weather conditioning:** add a parallel zero-shot weather classifier (sunny, rainy, snowy, overcast) run at index time, stored as image metadata. Weather-aware queries (“winter coat for a snowy commute”) could then filter or re-rank on this field directly, similar to how region-type filtering works today for garments.
- **Geotagged data:** if real-world deployment images carry EXIF/GPS metadata, city and climate context could be derived directly rather than inferred visually, giving a more reliable signal than zero-shot classification alone.

6.2 Improving precision

- **VLM reranking (Approach 5):** apply a small, free, locally-run VLM (e.g. Moondream2 or Qwen2-VL) to re-score only the top ~50 candidates returned by the fast vector search. Cost stays constant regardless of dataset size since only a bounded candidate set is touched per query, this directly targets the residual compositionality gap seen in query 1 (rank 9) and the pose-blindness seen in query 3.
- **Resolve the SigLIP-2 checkpoint issue** and re-benchmark against CLIP; SigLIP-2's sigmoid loss is expected to improve fine-grained color and attribute precision specifically.
- **NegCLIP / hard-negative fine-tuning:** fine-tune on attribute-swapped hard negatives (“red shirt blue pants” vs. the reverse) to improve the base embedding's inherent compositional understanding, reducing reliance on region-splitting alone.
- **Matryoshka (MRL) embeddings:** for scaling to larger datasets, adopt an MRL-capable embedding model (e.g. JinaCLIP-v2) to allow truncatable vectors, mitigating the ~4–5x storage growth from region-splitting discussed in Section 2.
- **Data sparsity fixes:** targeted supplementation for underrepresented garment categories (ties: 3 instances; cardigans: 10; jumpsuits: 18), mirroring the office/home scene supplementation already done for environment coverage.
- **Upgrade query decomposition** from rule-based keyword matching to a small local LLM call for structured extraction, more flexible to novel phrasing, still free and deterministic enough to run locally without an API dependency.

6.3 Scalability to 1 million images

The retrieval backbone (FAISS) scales by switching from a flat index to an approximate nearest-neighbor index (HNSW or IVF), which is a configuration change, not an architectural one, and remains sub-second at million-vector scale. VLM reranking stays cost-bounded regardless of dataset size, since it only touches a fixed top-k candidate set per query. The primary real cost at scale is the ~4–5x vector-count growth from region-splitting (garment crops in addition to whole-image vectors), which is a deliberate storage-for-accuracy tradeoff; MRL-based embedding truncation is the most direct mitigation.

7. Codebase

GitHub repository: <https://github.com/Srishti002/Multimodal-Fashion-Context-Retrieval>